



BIOL 603 - Final

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You are expected to work on your own and not with other students. You may use any and all class resources, readings, internet resources, etc. like you would for solving your own coding problems. Seeking examples of code from Stack Overflow, STHTDA, or coding blogs is fine, as these are tools you will continue to use as you analyze your own data. However, you MAY NOT use AI or any other code generation platform to determine what to do or write code for you.

The dataset you will be working with is a Rocky Mountain Biological Laboratory dataset that includes demography and spatial information for an alpine plant community in Colorado. You may choose to model any of these potential response variables.

Data Citation

Ray, Courtenay A.; Kapás, Rozália E.; Opedal, Øystein H.; Blonder, Benjamin W. (2022), Data from: Linking microenvironment modification to species interactions and demography in an alpine plant community, Dryad, Dataset, <https://doi.org/10.6078/D1342B>

Data product page: <https://datadryad.org/dataset/doi:10.6078/D1342B>

The README file for the dataset is included in the zip file for this final.

Final instructions

1. Create a new folder in your BIOL 603 folder on GitHub. Make sure I'm able to see it by checking the Collaborators tab in settings.
2. Create a new R script (or R notebook/Markdown document if you prefer), and read in the dataset.
3. Perform all necessary QAQC steps on this dataset, making sure to comment your code with brief explanations as to why you chose each step.

Remember that everyone approaches this a bit differently - I am looking for QAQC to be completed using the strategy that works best for you, not for it to be done in any specific order or using a specific tool. Walk me through your thought process so I can make sure that your technique and reasoning fit the requirements of the data.

4. Build summary statistics and create simple exploratory plots. Be sure to explain in brief comments why you chose the plots and statistics you did.

5. Determine which modeling approach is best suited to this data given the response variable you chose to model. Make sure to choose from among the models that have been covered in course material so far.
6. Check the assumptions of the model against the dataset to ensure your modeling approach will work. In comments, describe each assumption you must check, and the results of those checks.
7. Build the model you have determined is appropriate for the dataset. This may be an iterative model-selection process. Make sure to keep all code that you used to refine the model so that I can follow your thought process. Make sure to include brief comments at each step about what you have chosen to do and why.
8. Evaluate the model diagnostics, explaining in comments what they tell you about the fit of the model to the data.
9. Provide a detailed explanation in your own words about what each component of the model output is telling you. This is likely to come from `summary()` or `summary.glm()`.
10. Produce *two* refined, fully formatted figures showing the response variable in relation to all predictors. Use colors, shapes, etc. as needed, and make sure you are using a colorblind-safe color scheme. Use any type of plots you feel are appropriate for the dataset. Explain in brief comments what you chose and why.
11. Once you are happy with your completed script, push it to GitHub. ALSO submit it on Canvas as a backup!